# SCIENCE DISCOVERY LAB

# Basic 3 — Learner’s Guide

## 25 curriculum-aligned lessons • Observe • Explore • Explain • Practise

A colourful first science book about the natural world, materials, weather, water, energy, our bodies and safe everyday living.

Use this book with a teacher, parent or another trusted adult. All written explanations and questions are original educational material based on the uploaded NaCCA Basic 3 Science curriculum.

# Your discovery journey

## Diversity of Matter

1. Classifying Living Things • 2. Materials and Their Uses • 3. Objects Made from Several Materials • 4. Separate Solids from Liquids • 5. Changing States of Matter

## Cycles

6. Cycles in Daily Life • 7. Why the Sun Matters • 8. Rain, Snow, Hail and Sleet • 9. What Makes Water Impure? • 10. Air Is All Around Us • 11. Germinating Maize and Beans

## Systems

12. Body Parts Work Together • 13. Sun, Earth and Moon • 14. Homes for Living Things

## Forces and Energy

15. Light as Energy • 16. What Makes Heat? • 17. Everyday Uses of Heat • 18. Where Electricity Comes From • 19. Forces Make Things Move • 20. Care for Simple Machines

## Humans and the Environment

21. Keep Our Environment Clean • 22. Prevent Skin Problems • 23. Airborne Diseases • 24. How Food Gets Spoiled • 25. Keep the Air Clean

## Safety comes first

Do not touch broken glass, unknown substances, electrical sockets or hot objects. Never taste materials during experiments. Water, food and hygiene activities need appropriate adult guidance. During storms, stay in safe shelter.

# QUEST 1: Classifying Living Things

**Diversity of Matter | Classify plants and animals by observing the life processes they carry out.**

## LET’S LEARN

Plants and animals are both living things. Living things need nutrients and water, grow, respond to their surroundings, and reproduce. Green plants usually make their own food using sunlight. Animals must eat plants or other organisms. A bean plant grows roots and leaves; a goat moves to find food. Moving from place to place is not the only test of life: a growing tree stays rooted in one place.

## Science words to remember

Plants • Animals • Life processes • Classification




## Try it for yourself

Collect pictures of six living things. Sort them into plant and animal groups, and explain the life process shown by each picture.

Safety note: Work with a teacher or trusted adult. Use only safe everyday objects.

# QUEST 1: Show what you know

## Observe the four teaching steps

1. SORT: Display separate labelled cards for plants and animals.

2. OBSERVE: Show a young seedling changing as it grows.

3. COMPARE: Show an animal searching for food and a green plant in sunlight.

4. CLASSIFY: Move six picture cards into plant and animal groups.

## Guided practice — answer in your exercise book

1. What does a young mango tree belong to?
2. Which living thing usually makes food using sunlight?
3. Which group includes a goat?
4. Which living thing usually needs to eat other organisms?
5. What happens to a healthy bean seedling over time?
6. Which action shows an animal responding to danger?
7. Where should a chicken picture be placed?
8. How should grass be classified?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 2: Materials and Their Uses

**Diversity of Matter | Match everyday materials to products and explain how their properties make them useful.**

## LET’S LEARN

People choose materials for different jobs. Metal can be strong and hard, so some cooking pots are made of metal. Clear glass allows us to see through windows. Cotton makes comfortable, soft fabric. Wood is often used for furniture because it can be shaped and is fairly strong. Plastic can be moulded into bottles and buckets. Think about properties such as hardness, flexibility, transparency and absorbency before choosing a material.

## Science words to remember

Hardness • Flexibility • Transparency • Use




## Try it for yourself

Look at a clean plastic bottle, a metal spoon, paper and cotton cloth. Make a table with each material, its property and one appropriate use.

Safety note: Do not touch sharp glass or hot cooking pots.

# QUEST 2: Show what you know

## Observe the four teaching steps

1. MATCH: Show products next to candidate source materials.

2. TEST: Compare a hard metal spoon with a soft cotton cloth.

3. LOOK: Show a clear window compared with an opaque board.

4. CHOOSE: Match a bottle, chair, cloth and page to useful materials.

## Guided practice — answer in your exercise book

1. Why is metal useful for a cooking pot?
2. Which object is commonly made from strong wood?
3. Why can a flexible plastic strip bend?
4. Which material can be moulded into a bucket?
5. Which material lets light pass clearly through a window?
6. Which product needs a see-through material?
7. What is cotton cloth often used to make?
8. Which material is a sensible choice for a writing notebook page?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 3: Objects Made from Several Materials

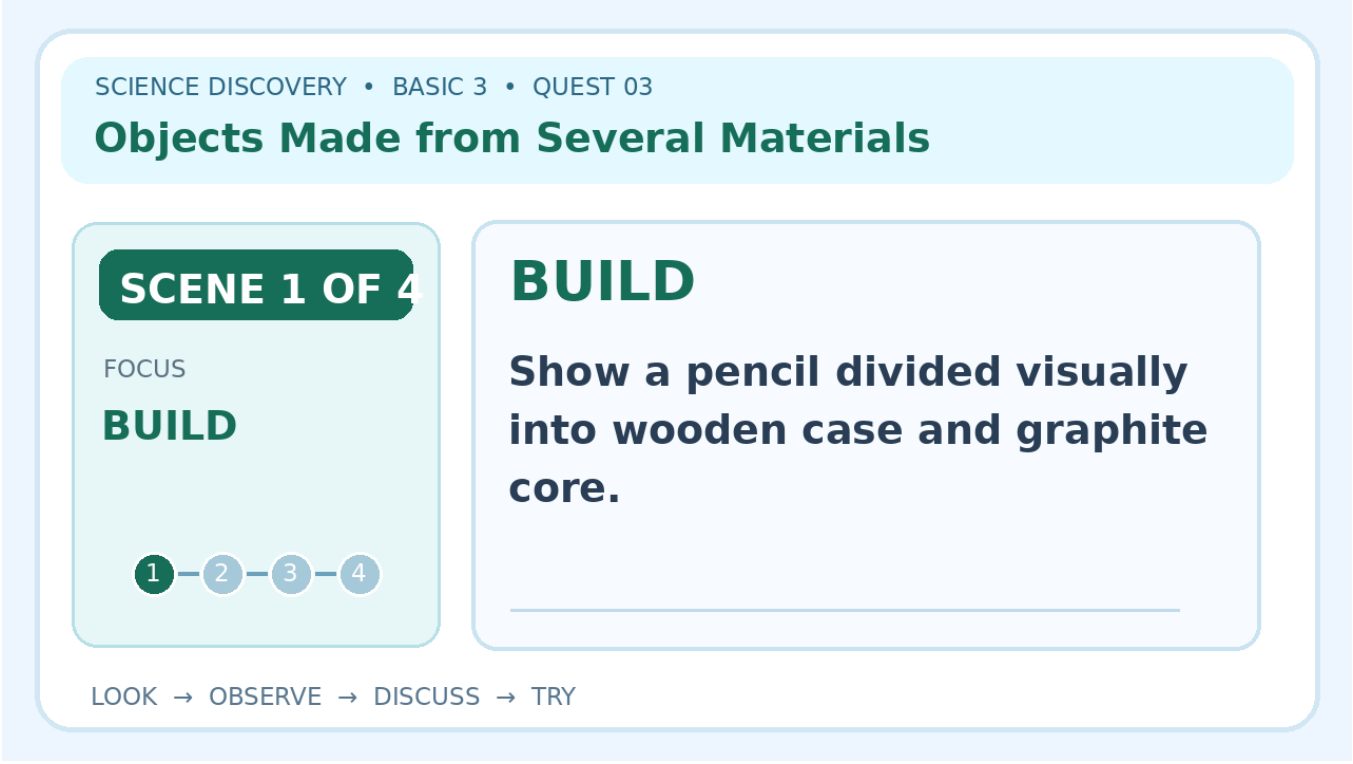
**Diversity of Matter | Identify more than one material in familiar objects and explain each material’s purpose.**

## LET’S LEARN

A single object may contain several materials. A pencil has wood around a graphite core, and some pencils also have a rubber eraser and metal holder. A mobile phone uses glass, metals and plastic. Shoes can combine fabric, rubber and other materials. The materials work together because each has a useful property. You can examine ordinary objects without dismantling or damaging them.

## Science words to remember

Components • Materials • Function • Safe inspection




## Try it for yourself

Choose an old shoe, pencil and torch. Draw their visible parts and label two different materials in each where possible.

Safety note: Work with a teacher or trusted adult. Use only safe everyday objects.

# QUEST 3: Show what you know

## Observe the four teaching steps

1. BUILD: Show a pencil divided visually into wooden case and graphite core.

2. LABEL: Highlight glass, plastic and metal parts of a phone.

3. MATCH: Show different materials in a shoe.

4. EXPLAIN: Connect each material with its job without opening the objects.

## Guided practice — answer in your exercise book

1. What two materials can be found in a pencil?
2. Which part of a pencil often leaves a dark mark?
3. Which material can form a phone screen?
4. Which material can make a shoe sole?
5. Why might a shoe use both fabric and rubber?
6. Why is wood placed around pencil graphite?
7. How should you inspect a phone’s materials?
8. What is the safest way to identify a toy’s parts?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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# QUEST 4: Separate Solids from Liquids

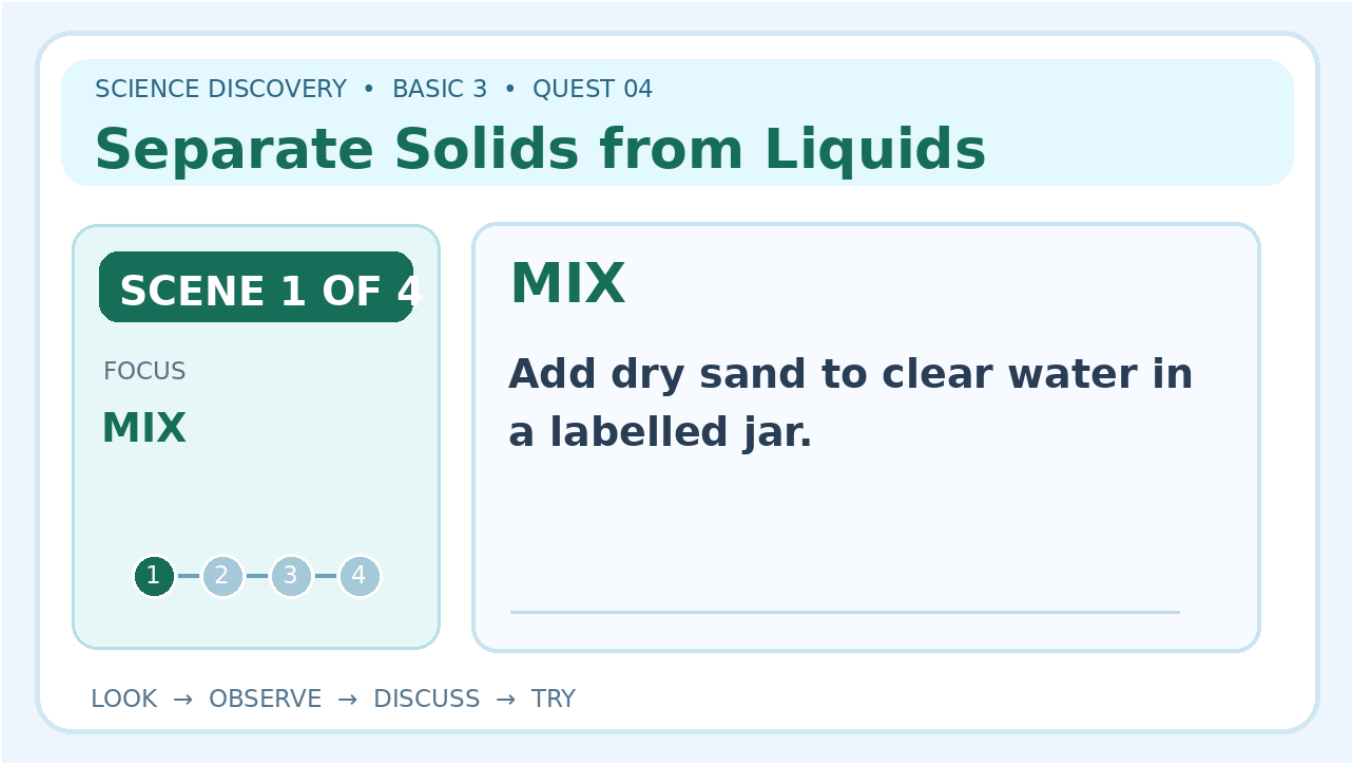
**Diversity of Matter | Describe a solid–liquid mixture and select a suitable separation method.**

## LET’S LEARN

When sand is mixed with water, the sand stays as a solid and the water remains a liquid. Many solid particles can be separated using a filter or by allowing them to settle, then carefully pouring off the liquid. A sieve can remove larger solid pieces, but it may not catch fine sand. Salt dissolves in water, so filtering does not remove dissolved salt; evaporating water is a different method that requires adult supervision. We must never drink untested separated water.

## Science words to remember

Mixture • Filtration • Settling • Dissolved solids




## Try it for yourself

Stand a clear jar of sand and water on a table. Observe settling, then let an adult help pour the clearer water carefully into another cup. Record what stayed behind.

Safety note: No tasting. Adults handle any heating or evaporation activity.

# QUEST 4: Show what you know

## Observe the four teaching steps

1. MIX: Add dry sand to clear water in a labelled jar.

2. SETTLE: Show sand particles collecting at the jar bottom.

3. POUR: Illustrate careful decanting while sediment remains.

4. FILTER: Contrast capturing undissolved sand with dissolved salt remaining in water.

## Guided practice — answer in your exercise book

1. What is sand stirred into water?
2. Which part of sandy water is liquid?
3. Which method can trap many undissolved fine particles?
4. What does a filter often keep behind from sandy water?
5. What happens when muddy water stands for a while?
6. Where does settled sand often collect?
7. Does filtering remove dissolved salt completely?
8. Which process can recover salt from saltwater with adult help?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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# QUEST 5: Changing States of Matter

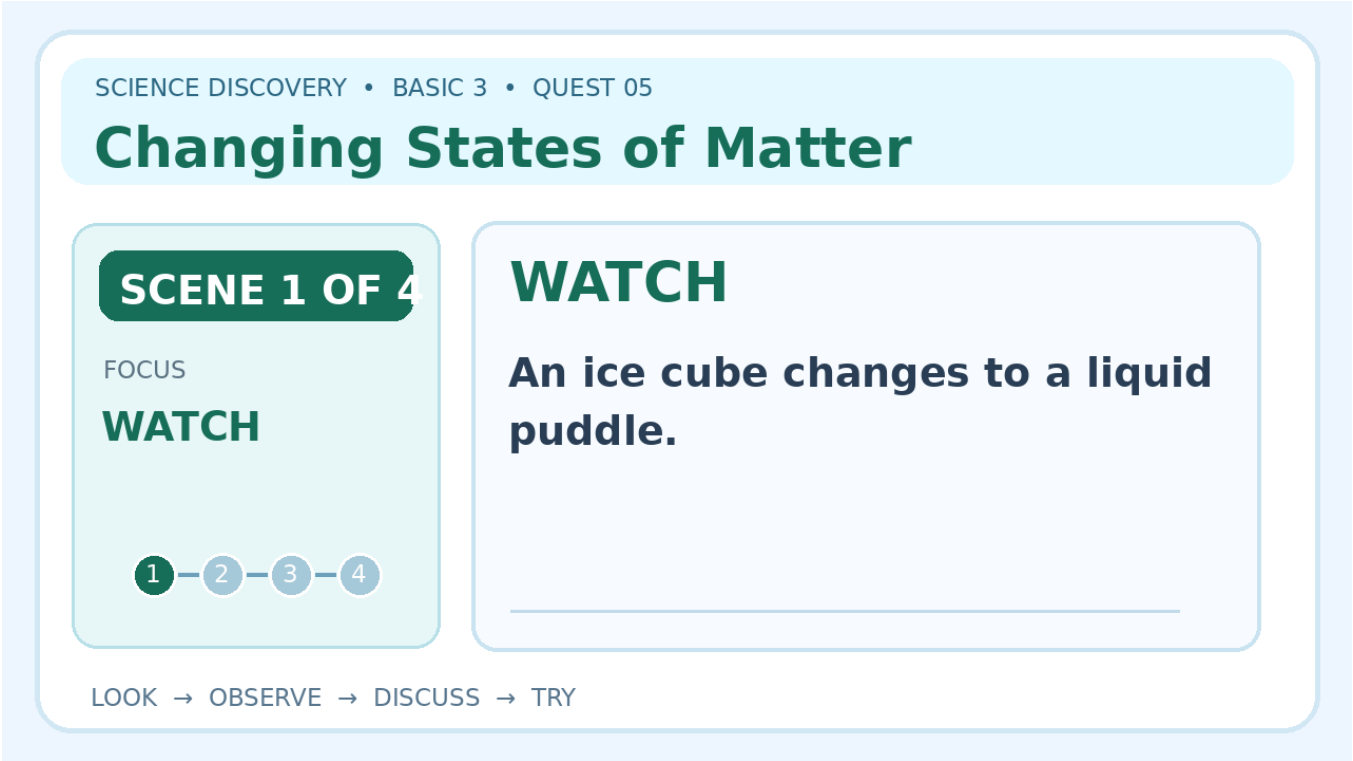
**Diversity of Matter | Explain melting, freezing, evaporation and condensation using water as an example.**

## LET’S LEARN

Water can be ice, liquid water or invisible water vapour. Ice melts when it gains enough heat, becoming liquid water. Water freezes when enough heat is removed. Liquid water can evaporate into water vapour. When warm moist air cools near a cold surface, water vapour can condense into tiny liquid droplets. In ordinary changes of state the substance remains water, even though its form changes.

## Science words to remember

Melting • Freezing • Evaporation • Condensation




## Try it for yourself

Watch an ice cube melt on a tray. Draw the starting ice and the water afterwards. Observe droplets outside a cold covered bottle without using hot water.

Safety note: Only observe cold-water and room-temperature examples; adults manage hot materials.

# QUEST 5: Show what you know

## Observe the four teaching steps

1. WATCH: An ice cube changes to a liquid puddle.

2. COOL: Show liquid water freezing into ice.

3. EVAPORATE: Show water becoming vapour above a wet surface.

4. CONDENSE: Show droplets appearing on the outside of a cold cup.

## Guided practice — answer in your exercise book

1. What happens when an ice cube gets warmer?
2. Which change turns solid ice into liquid?
3. What change makes liquid water into ice?
4. Which condition helps water freeze?
5. What happens to puddle water as it dries in sunshine?
6. Which state is formed when liquid water evaporates?
7. What makes small droplets appear outside a cold cup?
8. Which change turns water vapour into droplets?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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# QUEST 6: Cycles in Daily Life

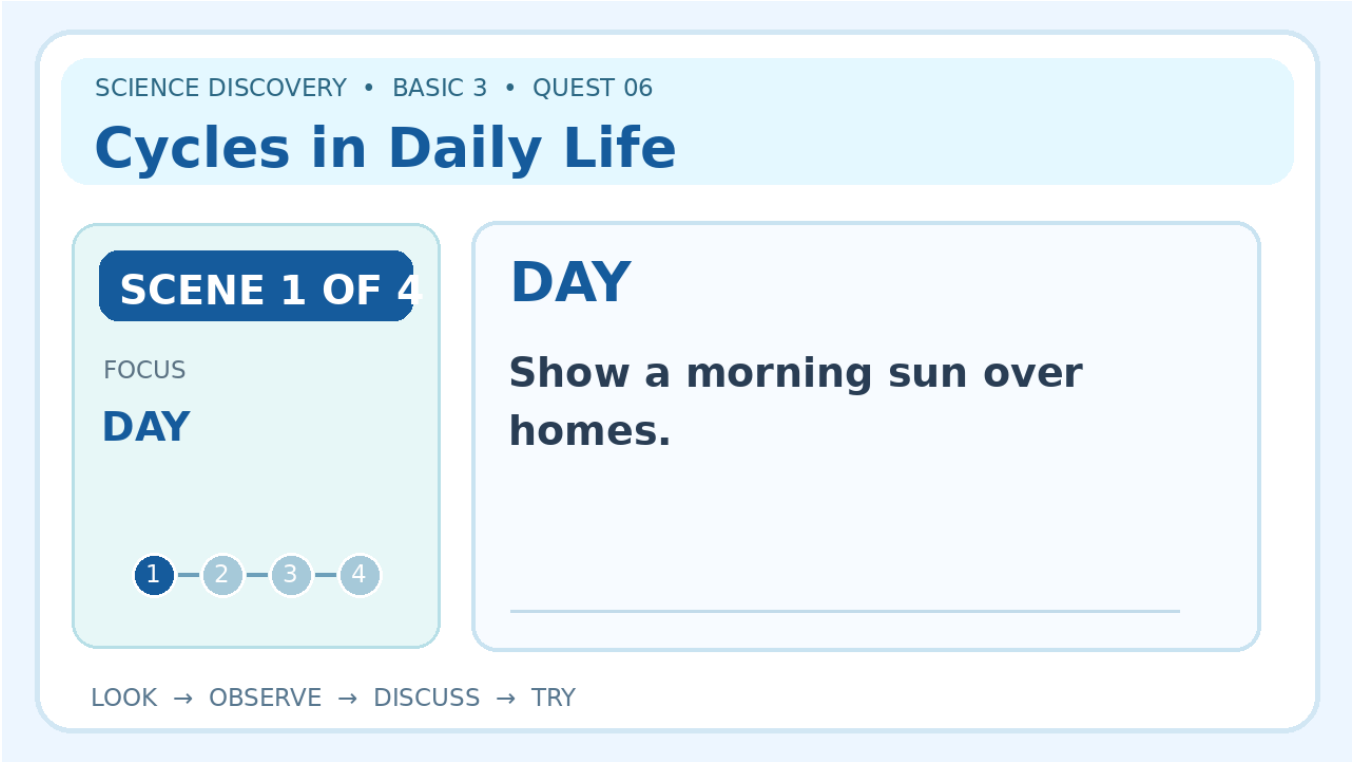
**Cycles | Recognise repeating events such as day and night, wet and dry seasons, and yearly celebrations.**

## LET’S LEARN

A cycle is a pattern of events that happens again and again. We see day followed by night, then day again. Ghana has periods of wetter and drier weather, though their timing differs between regions. School weeks and annual celebrations also repeat. Different cycles have different time intervals: day and night repeat in about one day, while an annual celebration happens once a year.

## Science words to remember

Daily cycle • Seasonal cycle • Annual events • Intervals




## Try it for yourself

Draw a circular arrow for day → night → day. Make a separate seasonal picture sequence from observations in your own community.

Safety note: Work with a teacher or trusted adult. Use only safe everyday objects.

# QUEST 6: Show what you know

## Observe the four teaching steps

1. DAY: Show a morning sun over homes.

2. NIGHT: Show the same homes under the moon and stars.

3. SEASON: Compare a wetter period with a drier period in Ghana.

4. CYCLE: Draw arrows linking the repeating order and different intervals.

## Guided practice — answer in your exercise book

1. Which event repeats after each night?
2. What repeats in a daily pattern?
3. Which pair describes broad seasonal weather patterns in Ghana?
4. Which observation suggests a wetter period?
5. Which celebration comes once each year?
6. Which cycle is annual?
7. About how long is one day–night cycle?
8. How often does an annual event recur?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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# QUEST 7: Why the Sun Matters

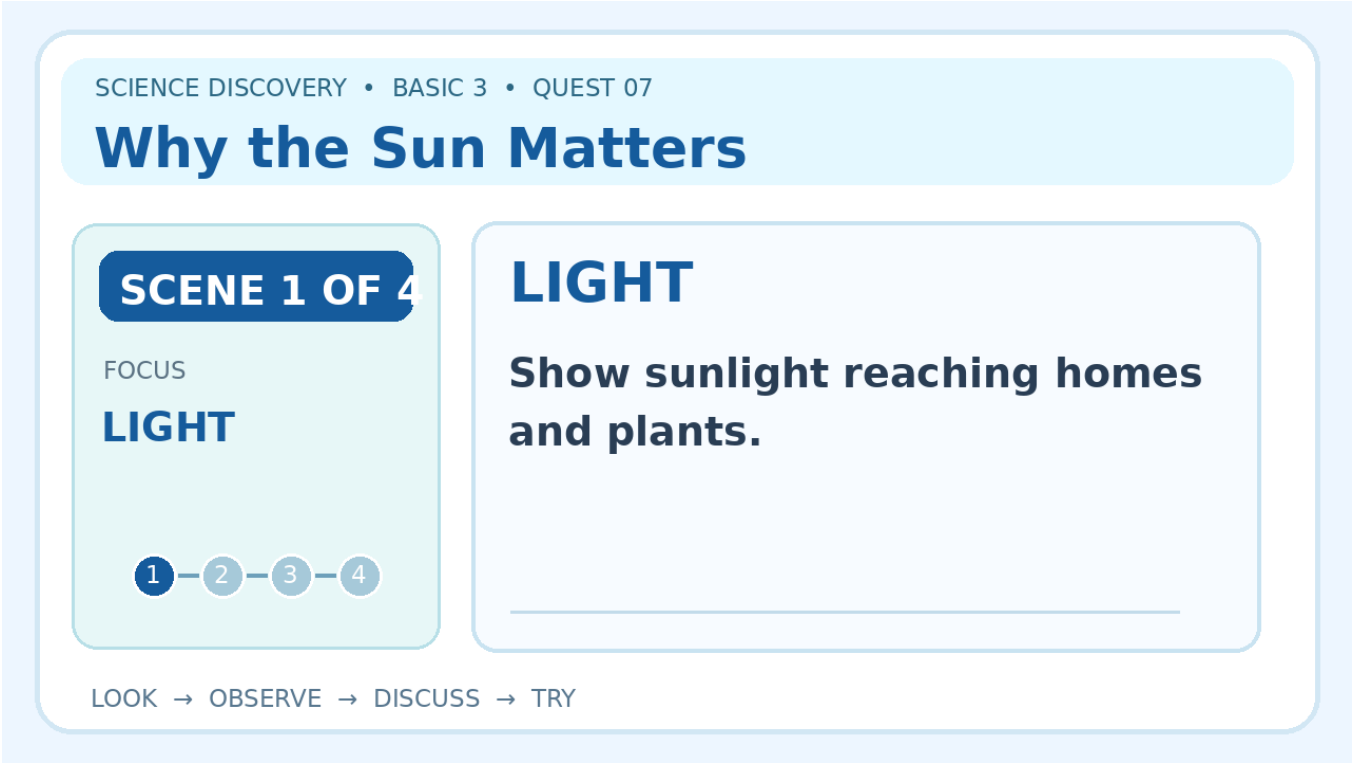
**Cycles | Explain ways the Sun benefits Earth, plants, animals and people.**

## LET’S LEARN

The Sun gives Earth light and warmth. Sunlight helps green plants make food. Sunshine can help dry washed clothes and harvested grains. The Sun’s energy also helps move water through evaporation in the water cycle. Spending too long in strong midday sun without protection can be harmful, so we should find shade, drink water and protect our skin.

## Science words to remember

Light • Warmth • Plants • Drying




## Try it for yourself

Compare one damp cloth dried in shade with another in a suitable sunny place, with adult support. Predict which may dry faster and record what you notice.

Safety note: Never look directly at the Sun; protect skin and avoid heat exhaustion.

# QUEST 7: Show what you know

## Observe the four teaching steps

1. LIGHT: Show sunlight reaching homes and plants.

2. WARM: Show gentle warmth reaching the ground.

3. GROW: Show a green plant using light to make food.

4. DRY: Show wet clothes drying outdoors, with a shaded rest area.

## Guided practice — answer in your exercise book

1. What does the Sun supply during daytime?
2. Why is daylight helpful in a classroom?
3. Which kind of energy from the Sun warms Earth?
4. What should you do when the midday sun feels too hot?
5. Why do green plants need sunlight?
6. Which living thing often uses sunlight directly to make food?
7. What can sunlight help dry on a line?
8. Which activity can benefit from sunshine?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 8: Rain, Snow, Hail and Sleet

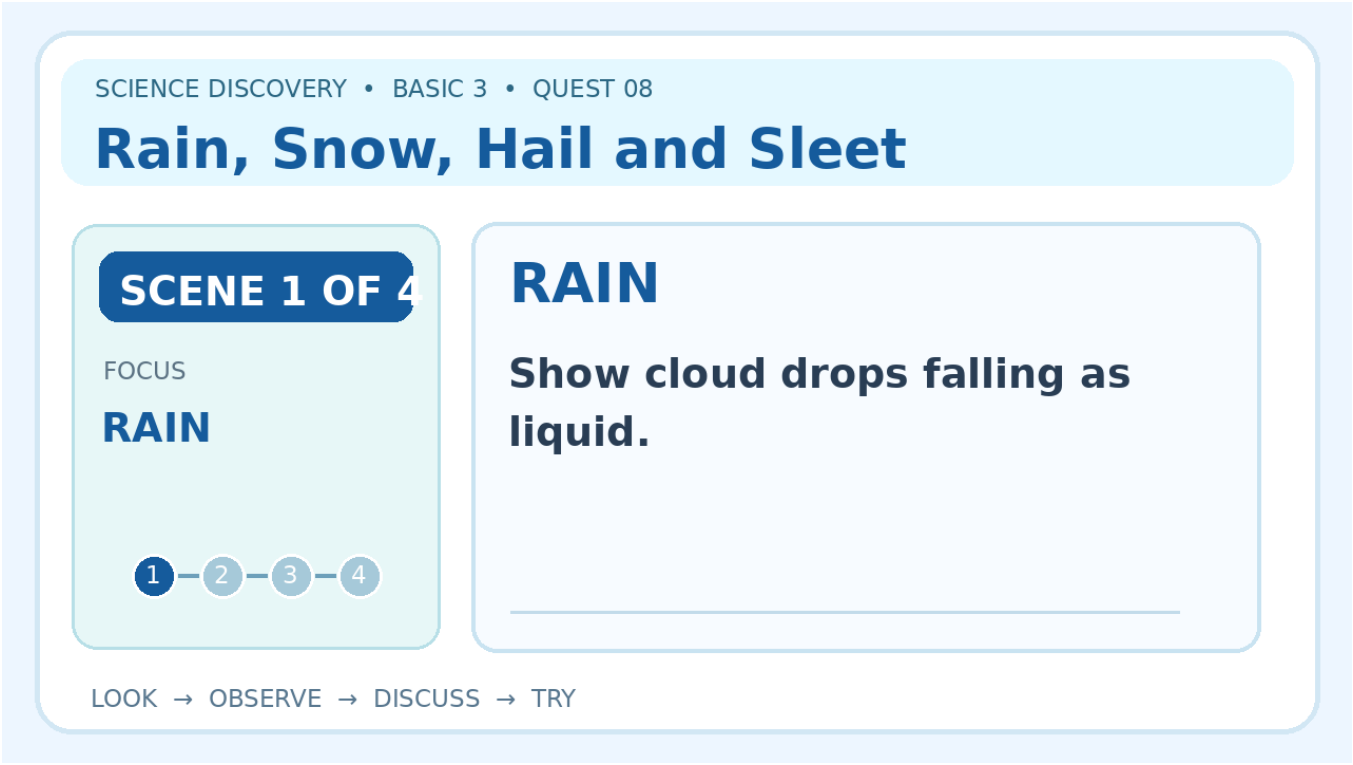
**Cycles | Identify four types of precipitation and compare how they reach the ground.**

## LET'S LEARN

Precipitation is water that falls from clouds. Rain falls mainly as liquid drops. Snow falls as ice crystals that can collect as soft flakes in cold weather. Hail is made of lumps or balls of ice formed within strong storm clouds. Sleet can mean ice pellets in some countries; the term is used differently in some places, but here it refers to small ice pellets. Rain is familiar in Ghana; snow, hail and sleet may be better observed through pictures.

## Science words to remember

Rain • Snow • Hail • Sleet




## Try it for yourself

Sort four weather picture cards into rain, snow, hail and sleet. Explain whether each reaches the ground mostly as liquid water or ice.

Safety note: Do not go outdoors to collect hail during a storm; use pictures instead.

# QUEST 8: Show what you know

## Observe the four teaching steps

1. RAIN: Show cloud drops falling as liquid.

2. SNOW: Show tiny ice crystals collecting into snowflakes.

3. HAIL: Show hard rounded ice particles from a storm cloud.

4. SLEET: Show small ice pellets and compare the four forms.

## Guided practice — answer in your exercise book

1. Which precipitation falls mainly as liquid droplets?
2. What commonly falls during a Ghanaian rainstorm?
3. Which precipitation contains delicate ice crystals?
4. Which kind may form soft flakes in very cold air?
5. Which precipitation contains rounded pieces of ice from strong storms?
6. What can be dangerous when it falls as large hard ice balls?
7. What is sleet in this lesson?
8. Which is a form of frozen precipitation rather than liquid rain?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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# QUEST 9: What Makes Water Impure?

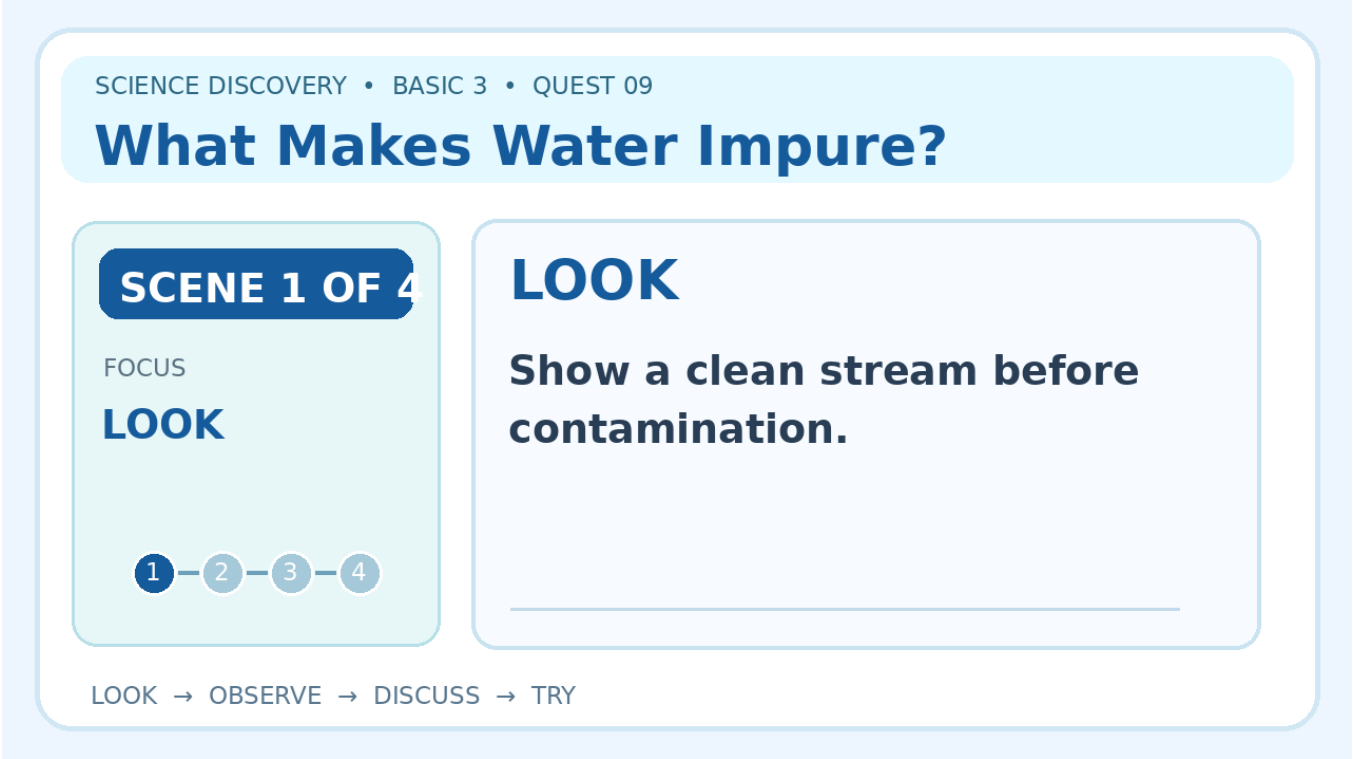
**Cycles | Identify several sources of water pollution and explain how to keep water sources safer.**

## LET’S LEARN

Water may become impure when waste, dirty floodwater, household chemicals, mining waste, pesticides or animal droppings enter it. Clear water is not automatically safe to drink; harmful contaminants may be invisible. People can protect water by disposing of waste properly, keeping chemicals away from streams and using treated drinking water. Report polluted streams and drainage areas to a responsible adult.

## Science words to remember

Litter • Chemicals • Floodwater • Prevention




## Try it for yourself

Draw a stream with two labelled panels: polluted by rubbish and protected by clean surroundings. Discuss three actions that stop contamination.

Safety note: Do not touch polluted water or unknown chemicals; never taste samples.

# QUEST 9: Show what you know

## Observe the four teaching steps

1. LOOK: Show a clean stream before contamination.

2. ADD: Show waste and runoff moving toward water, clearly labelled unsafe.

3. TRACE: Use arrows to show how floodwater can carry dirt.

4. PROTECT: Show covered bins and safe treated drinking water.

## Guided practice — answer in your exercise book

1. What can pollute a stream near a market?
2. Which action makes a pond dirtier?
3. Which substance should stay out of drinking water?
4. What may pollute a stream near a mining area?
5. How can flooding affect a community water source?
6. What might rainwater carry from a dirty drain?
7. Which action helps protect a well?
8. How should drinking water be chosen?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 10: Air Is All Around Us

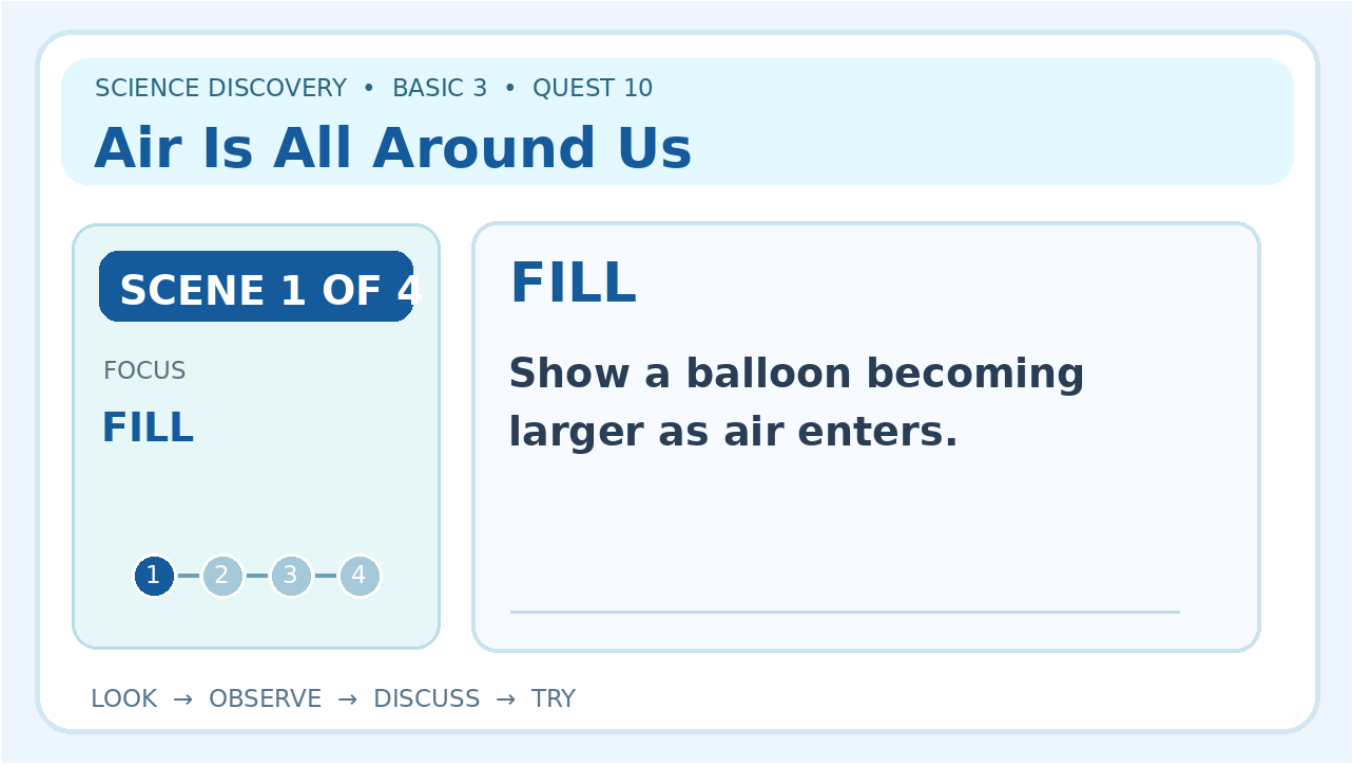
**Cycles | Demonstrate that air occupies space, has mass and can push or move objects.**

## LET’S LEARN

Air is a mixture of gases. We cannot usually see it, but it is all around us. A balloon expands when air fills it, showing that air occupies space. Air has mass even though a small amount is light. Moving air is wind, which can turn pinwheels or move leaves. People and other animals need oxygen from air to live. We can make a room more comfortable by allowing fresh air to circulate.

## Science words to remember

Space • Mass • Wind • Breathing




## Try it for yourself

Inflate a balloon with adult help, then deflate it and compare its size. Observe a paper pinwheel moving in a light breeze.

Safety note: Do not put balloons over heads or faces. Keep broken balloon pieces from young children.

# QUEST 10: Show what you know

## Observe the four teaching steps

1. FILL: Show a balloon becoming larger as air enters.

2. WEIGH: Illustrate air having small but real mass.

3. MOVE: Show a pinwheel rotating in moving air.

4. BREATHE: Show fresh-air ventilation in a classroom.

## Guided practice — answer in your exercise book

1. Why does a balloon expand when inflated?
2. What fills the inside of a blown-up balloon?
3. Does air have mass?
4. What is true about an inflated balloon compared with an empty one?
5. What do we call air that is moving?
6. What can a gentle breeze move?
7. Which gas in air do humans need for breathing?
8. How can a classroom get more fresh air safely?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 11: Germinating Maize and Beans

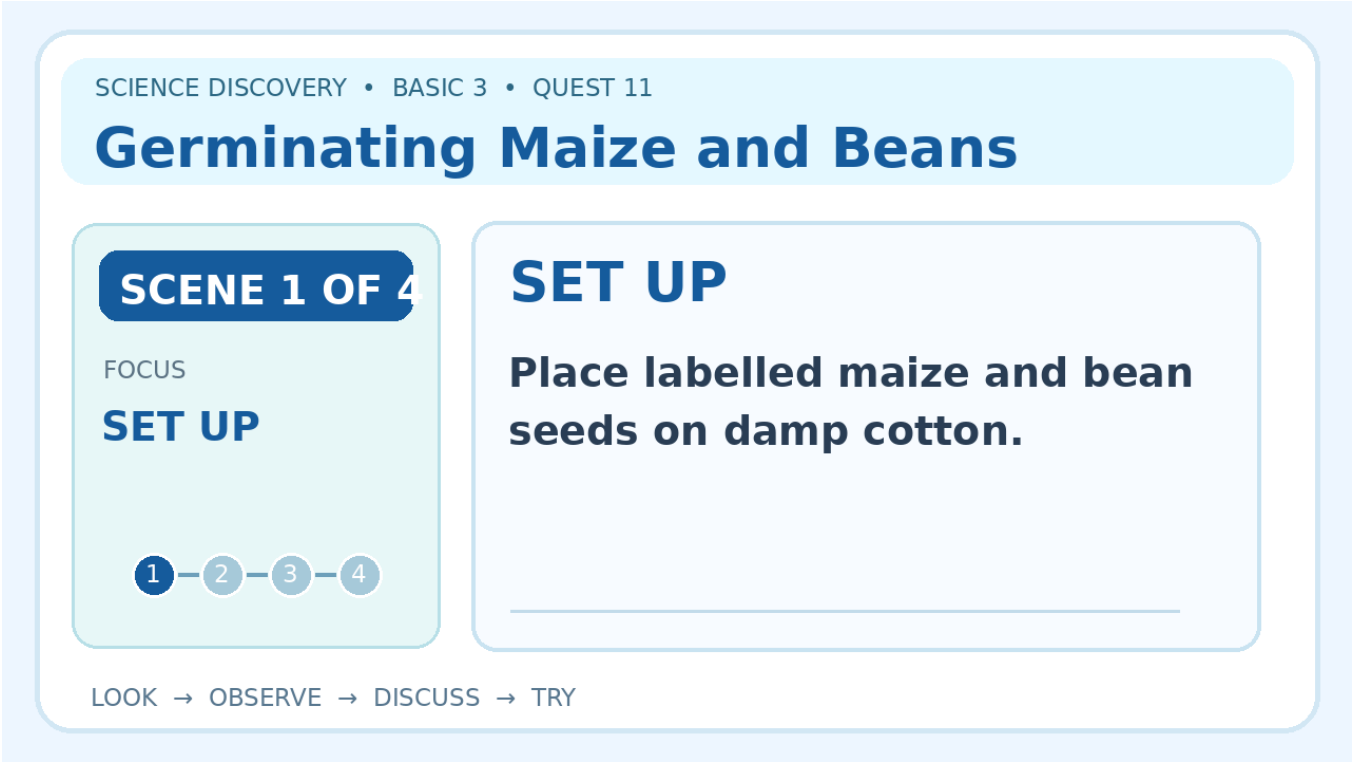
**Cycles | Observe seed germination and investigate how moisture affects the sprouting of maize and bean seeds.**

## LET’S LEARN

A viable seed can begin to grow into a new plant when conditions are suitable. Water is needed for germination; seeds also need oxygen and an appropriate temperature. A maize seed and a bean seed can sprout on damp cotton wool without soil for a short observation. Compare a damp container with a dry one while keeping other conditions similar. Record changes daily and do not assume both kinds sprout at the same speed.

## Science words to remember

Damp cotton • Dry control • Root and shoot • Fair test




## Try it for yourself

Place maize and bean seeds on damp cotton in transparent containers. Set up a matching dry-cotton comparison. Record the presence of roots and shoots over several days with adult help.

Safety note: Wash hands after handling seeds; do not eat or taste experiment seeds.

# QUEST 11: Show what you know

## Observe the four teaching steps

1. SET UP: Place labelled maize and bean seeds on damp cotton.

2. COMPARE: Show a second labelled container with dry cotton.

3. OBSERVE: Depict root and shoot appearing after suitable days.

4. RECORD: Put daily observations in two side-by-side charts.

## Guided practice — answer in your exercise book

1. Where might seeds begin sprouting without soil?
2. What should you use to moisten cotton for a germination test?
3. What often happens to seeds on completely dry cotton?
4. Why include a dry-cotton container?
5. Which part often appears first when a seed sprouts?
6. What grows upward from a sprouting seed?
7. What should remain similar in two seed containers?
8. What is the best way to compare sprouting over time?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 12: Body Parts Work Together

**Systems | Explain how eyes, ears, legs, hands and other external body parts cooperate in activities.**

## LET’S LEARN

Our body parts work together to perform everyday tasks. During a football game, eyes track the ball, legs move, and feet kick. When a loud horn is heard, ears detect the sound, eyes locate where it came from and legs can move the body to safety. Hands and eyes cooperate when writing. Each external body part is important, and we should keep our bodies clean and healthy.

## Science words to remember

Eyes • Ears • Limbs • Cooperation




## Try it for yourself

With a teacher, safely roll a soft ball to a partner. Discuss how eyes, hands and feet help with watching, receiving and moving it.

Safety note: Only use a soft ball, clear space and adult supervision; no dangerous running games.

# QUEST 12: Show what you know

## Observe the four teaching steps

1. SEE: Highlight the eyes watching a ball.

2. HEAR: Show ears detecting a bell or voice.

3. MOVE: Illustrate legs moving and hands catching.

4. COOPERATE: Connect eyes, ears and limbs in a safe reaction.

## Guided practice — answer in your exercise book

1. Which part helps you see a ball flying?
2. What helps you read letters on a page?
3. Which part notices a bicycle bell?
4. How do you notice someone calling your name?
5. Which parts help you run to safety?
6. Which parts can help you catch a soft ball?
7. Which body parts work together when handwriting?
8. Which combination helps you respond to a nearby vehicle?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 13: Sun, Earth and Moon

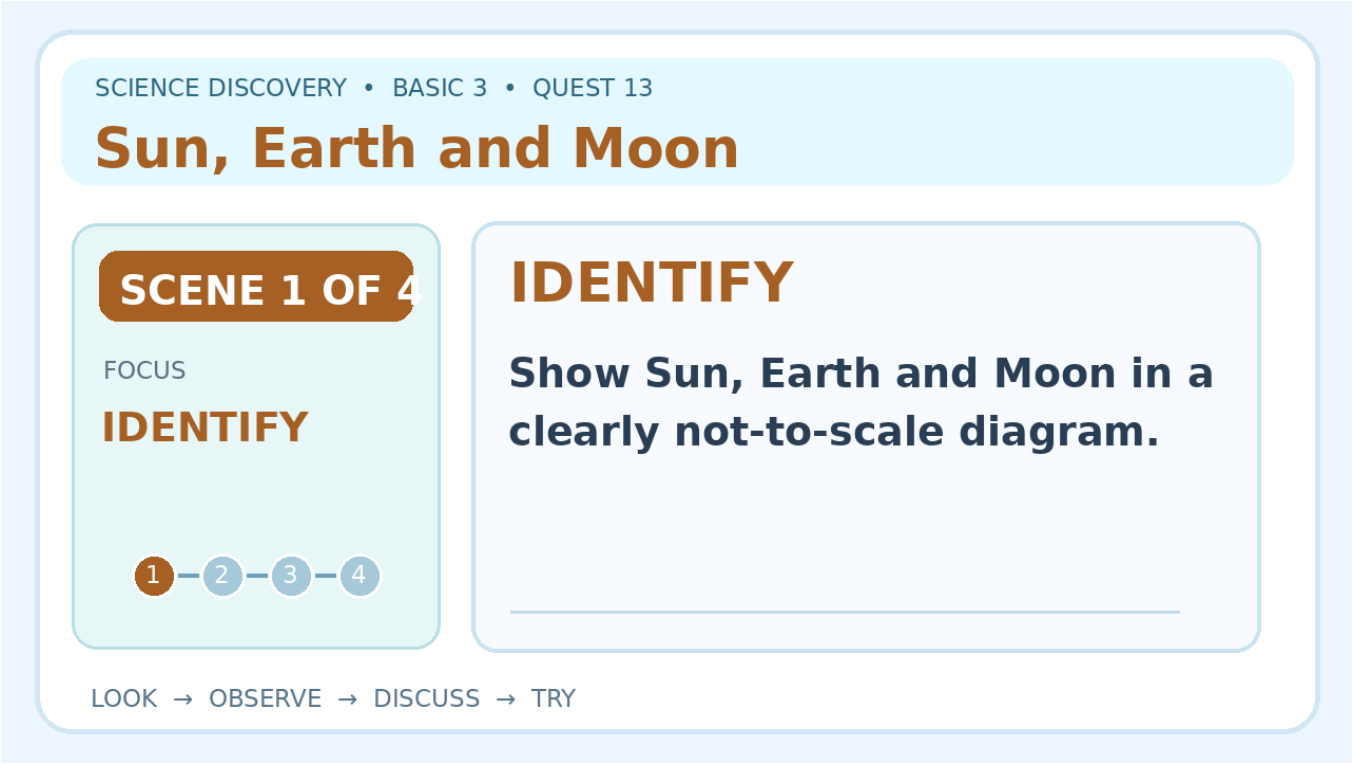
**Systems | Identify the Sun, Earth and Moon and describe how Earth and Moon move.**

## LET’S LEARN

The Sun is the star at the centre of our solar system. Earth is a planet that travels around the Sun. The Moon is Earth’s natural satellite and travels around Earth. Earth also rotates, giving rise to day and night. Sunlight reaches Earth and makes the Moon appear bright when the Moon reflects it. The Moon does not make its own sunlight. Classroom models are not drawn to actual distance or size.

## Science words to remember

Sun • Earth • Moon • Motion




## Try it for yourself

Use a large ball as the Sun, a medium ball as Earth and a smaller ball as Moon. Carefully model Earth going around the Sun and Moon going around Earth.

Safety note: Never look directly at the Sun; classroom models are safe substitutes.

# QUEST 13: Show what you know

## Observe the four teaching steps

1. IDENTIFY: Show Sun, Earth and Moon in a clearly not-to-scale diagram.

2. ORBIT: Demonstrate Earth travelling around the Sun.

3. ORBIT: Demonstrate Moon travelling around Earth.

4. ROTATE: Show Earth turning to explain day and night.

## Guided practice — answer in your exercise book

1. Which object is a star?
2. What provides most natural daylight on Earth?
3. What kind of object is Earth?
4. What does Earth travel around?
5. Which object is Earth’s natural satellite?
6. Where does moonlight mainly come from?
7. What movement contributes to day and night?
8. What travels around Earth?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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# QUEST 14: Homes for Living Things

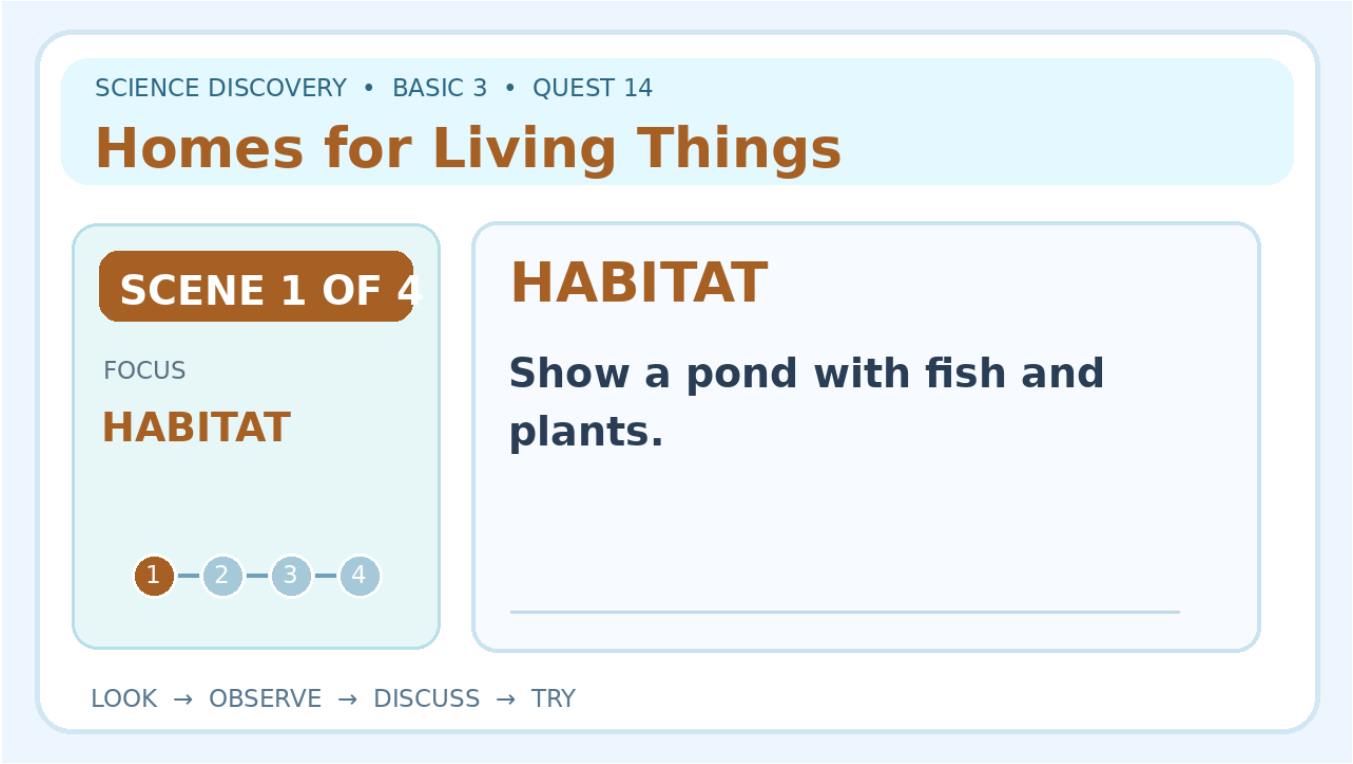
**Systems | Identify common habitats and explain how an organism’s body helps it live there.**

## LET’S LEARN

A habitat is the natural place where an organism lives. Fish have fins that help them swim in ponds or rivers. Birds have wings, and many build nests in trees. Grasshoppers live in grassy areas where they can find food and shelter. A frog can live both in water and on land, while most fish cannot breathe safely out of water. Living things depend on suitable water, air, food and shelter.

## Science words to remember

Water habitat • Tree habitat • Grass habitat • Adaptations




## Try it for yourself

Observe a school garden from a safe distance and draw the plants and animals seen there. Match birds to trees and fish to water, with a reason.

Safety note: Observe animals from a distance. Do not disturb nests or enter ponds.

# QUEST 14: Show what you know

## Observe the four teaching steps

1. HABITAT: Show a pond with fish and plants.

2. HABITAT: Show birds living among tree branches.

3. HABITAT: Show insects in grass.

4. MATCH: Connect fins, wings and legs to living in suitable places.

## Guided practice — answer in your exercise book

1. Where do many tilapia fish live?
2. What helps a fish swim?
3. Where might you find a bird’s nest?
4. What feature helps many birds fly?
5. Where is a grasshopper likely to find food?
6. Which habitat suits a grasshopper best?
7. Why do fins help a fish?
8. Why does a habitat matter?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 15: Light as Energy

**Forces and Energy | Explain how light helps us see and identify safe sources of light.**

## LET’S LEARN

Light is a form of energy that helps our eyes see objects. The Sun is a natural light source. Electric lamps and torches are artificial sources. An object becomes visible when light from a source enters our eyes after reflecting from the object, or comes from the glowing object itself. Light cannot pass through every material: an opaque cover blocks it. In dark places we can use a torch, but we must not point bright lights into eyes.

## Science words to remember

Sunlight • Torches • Seeing • Shadow




## Try it for yourself

Darken a corner safely, then use a battery torch to reveal a book. Place cardboard between the torch and the book to notice how the beam is blocked.

Safety note: Use battery torches; no flames, exposed wires or direct bright light into eyes.

# QUEST 15: Show what you know

## Observe the four teaching steps

1. SOURCE: Show daylight coming from the Sun.

2. SOURCE: Show a battery torch lighting a book.

3. SEE: Trace light to the book and then to a learner’s eyes.

4. SHADOW: Show opaque cardboard blocking light to form a shadow.

## Guided practice — answer in your exercise book

1. Which is a natural source of light?
2. What lights the Earth during the day?
3. Which item uses batteries to give light?
4. What is an artificial light source?
5. Why can we see a book in good light?
6. What helps people read at night?
7. What can an opaque piece of cardboard form when it blocks light?
8. Where does a shadow form?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 16: What Makes Heat?

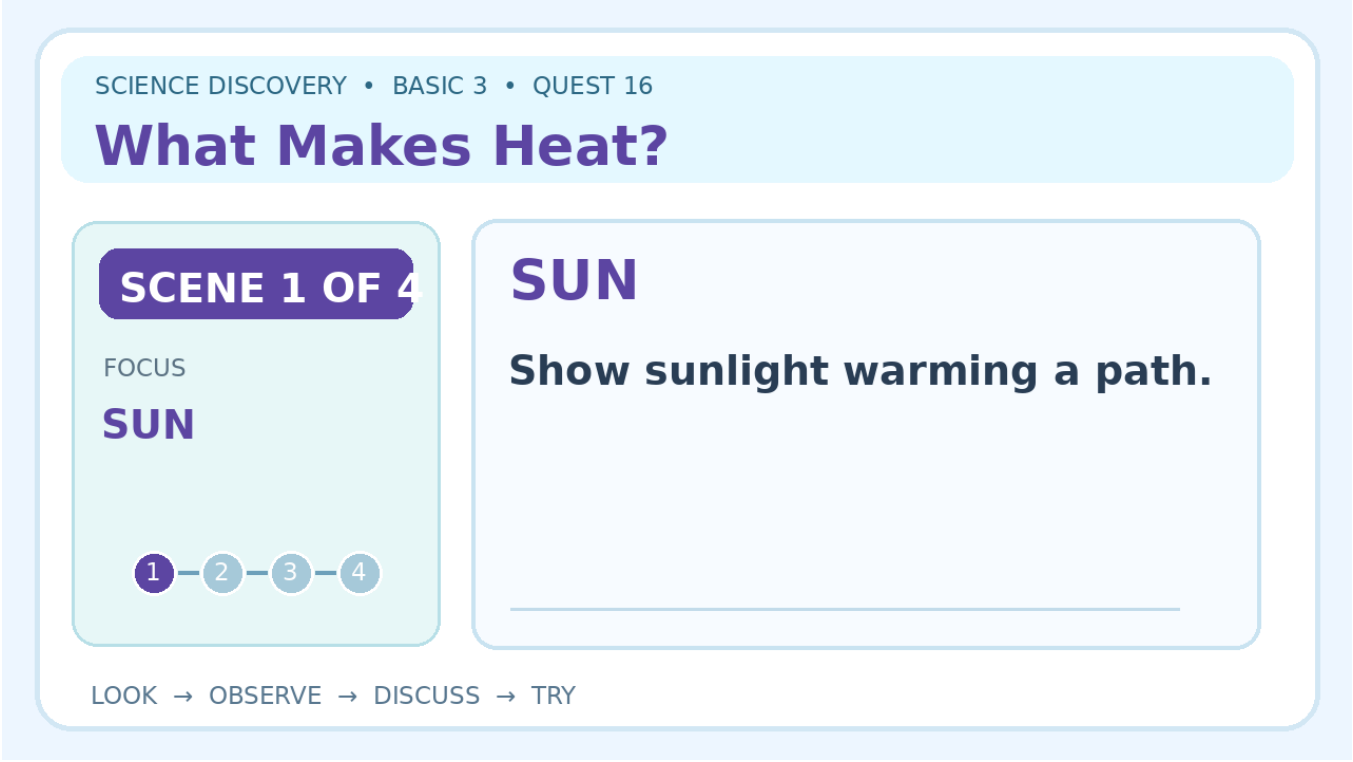
**Forces and Energy | Identify heat as energy and name natural and everyday sources of heat.**

## LET’S LEARN

Heat is energy that moves from warmer things to cooler things. The Sun warms the Earth. A stove can heat food, and rubbing hands together can make palms feel warmer because of friction. Some electrical appliances also give off heat. Hot objects can burn skin even when they do not look bright or red. We should observe heat demonstrations from a safe distance and ask adults to handle hot equipment.

## Science words to remember

Sun • Friction • Appliances • Safety




## Try it for yourself

Rub clean palms together gently for a few seconds and notice whether they feel warmer. Compare this with standing in shade versus sunlight, without staying in heat too long.

Safety note: Children must not light fires or operate hot appliances.

# QUEST 16: Show what you know

## Observe the four teaching steps

1. SUN: Show sunlight warming a path.

2. RUB: Illustrate palms rubbing gently to create warmth.

3. APPLIANCE: Show a stove producing heat for cooking, adult only.

4. SAFETY: Show a child keeping away from a hot surface.

## Guided practice — answer in your exercise book

1. Which natural object can make the ground warm?
2. What warms a sunny pavement?
3. What can warm hands when they are rubbed together?
4. Which activity uses friction to create warmth?
5. Which appliance is meant to provide heat for cooking?
6. Which device may become warm when used?
7. What should you do before touching an unknown hot pot?
8. Why are hot objects handled carefully?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 17: Everyday Uses of Heat

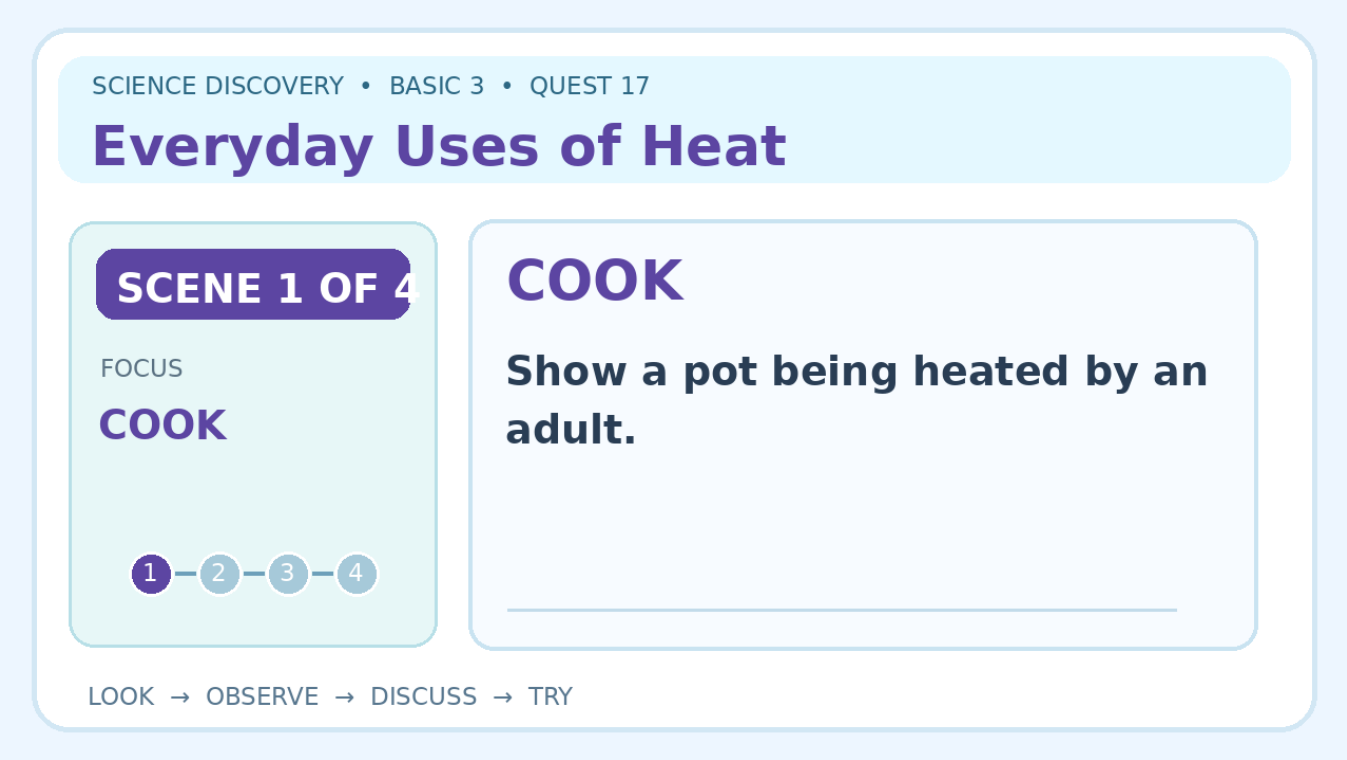
**Forces and Energy | Describe safe everyday uses of heat in homes, farms and workplaces.**

## LET’S LEARN

People use heat for cooking, boiling water, ironing clothes, drying and some food preservation. A baker uses an oven, a tailor may use an electric iron, and a blacksmith uses high temperatures to shape metal. Farmers can dry some crops in appropriate sunlight. Heat is useful, but burns and fires are hazards. Children should observe heat-related work without touching hot tools.

## Science words to remember

Cooking • Drying • Ironing • Jobs




## Try it for yourself

Create picture cards of a baker, tailor, farmer and cook. Match each worker to a safe explanation of why heat is useful in their work.

Safety note: Hot cooking, ironing and metalwork demonstrations are for adults only.

# QUEST 17: Show what you know

## Observe the four teaching steps

1. COOK: Show a pot being heated by an adult.

2. DRY: Show harvested maize drying in a suitable clean place.

3. IRON: Show a tailor using an electric iron safely.

4. WORK: Match baker and blacksmith to their uses of heat.

## Guided practice — answer in your exercise book

1. Why does a cook need heat?
2. Which activity commonly uses a stove?
3. How can sunshine help harvested grains?
4. Which worker may dry crops in sunshine?
5. What appliance helps remove creases from clothes?
6. Which work often uses a hot iron?
7. Who may use high heat to shape metal?
8. Which worker commonly uses a heated oven?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 18: Where Electricity Comes From

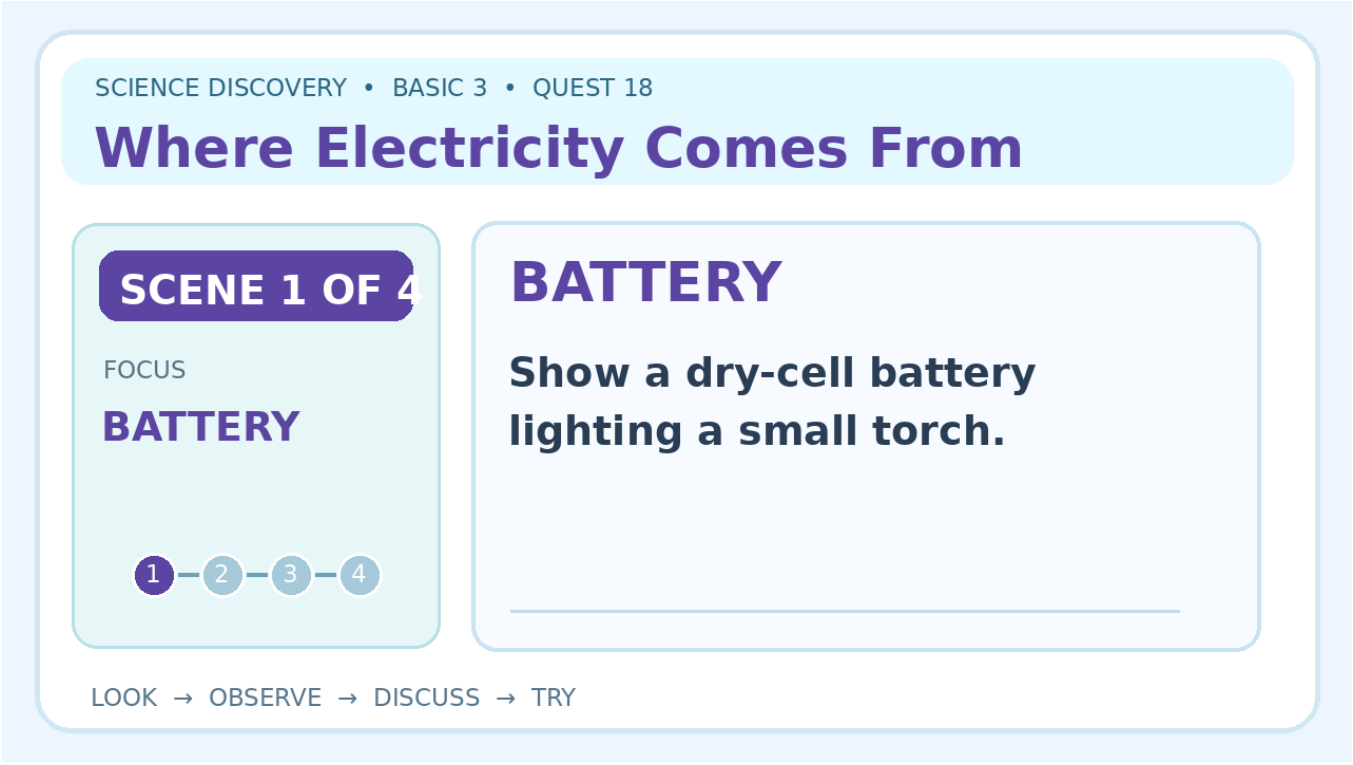
**Forces and Energy | Identify batteries, solar panels, moving water and power stations as sources of electrical energy.**

## LET’S LEARN

Electricity can come from several sources. A dry-cell battery stores energy used by a torch. Solar panels can convert energy from sunlight into electricity. Moving water can turn turbines at hydroelectric stations. Thermal power stations use heat energy to generate electricity. Ghana has hydroelectric facilities at Akosombo and Bui. Electricity is useful but dangerous if handled unsafely; children must never experiment with mains sockets or power lines.

## Science words to remember

Batteries • Solar • Hydropower • Electrical safety




## Try it for yourself

Observe a battery-powered torch and a picture of a solar panel, water turbine and power station. Create a four-column source chart.

Safety note: Never open sockets, handle mains wiring or touch fallen electrical lines.

# QUEST 18: Show what you know

## Observe the four teaching steps

1. BATTERY: Show a dry-cell battery lighting a small torch.

2. SOLAR: Show sunlight reaching a solar panel.

3. HYDRO: Show flowing water turning a turbine at a dam.

4. SAFETY: Show safe distance from wall sockets and cables.

## Guided practice — answer in your exercise book

1. What supplies a small torch with electrical energy?
2. Which object stores energy for a battery toy?
3. What converts sunlight into electricity?
4. Which source relies on sunlight?
5. How can falling or moving water help make electricity?
6. What energy source is used at a hydroelectric station?
7. Which is safe for a child to explore under supervision?
8. What should you do near a fallen power cable?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 19: Forces Make Things Move

**Forces and Energy | Explain that pushes, pulls and some non-contact forces can change how objects move.**

## LET’S LEARN

A force is a push or a pull. You can push a toy car to start it moving, pull a small wagon toward you, or stop a ball by gently catching it. Magnets may attract some iron objects without touching them. Rubbing certain plastic objects can cause small pieces of paper to move because of static electricity. We must always choose safe, lightweight objects for classroom demonstrations.

## Science words to remember

Push • Pull • Magnetism • Motion




## Try it for yourself

Push a toy car gently on a smooth floor and then stop it by hand. Observe a magnet attracting a paper clip, while an adult supervises.

Safety note: Use lightweight toys only; do not throw stones or use sharp metal pins.

# QUEST 19: Show what you know

## Observe the four teaching steps

1. PUSH: Show a finger starting a toy car moving.

2. PULL: Show a string drawing a small wagon nearer.

3. MAGNET: Show an iron clip moving toward a magnet.

4. STOP: Show hands slowing a soft rolling ball to rest.

## Guided practice — answer in your exercise book

1. What force moves a toy car away from your hand?
2. Which action is a push?
3. Which action is a pull?
4. What kind of force acts when you draw a small wagon toward you?
5. What can attract an iron paper clip without contact?
6. Which material is attracted strongly by a typical classroom magnet?
7. What may happen when you catch a rolling ball?
8. What can a force change about an object?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 20: Care for Simple Machines

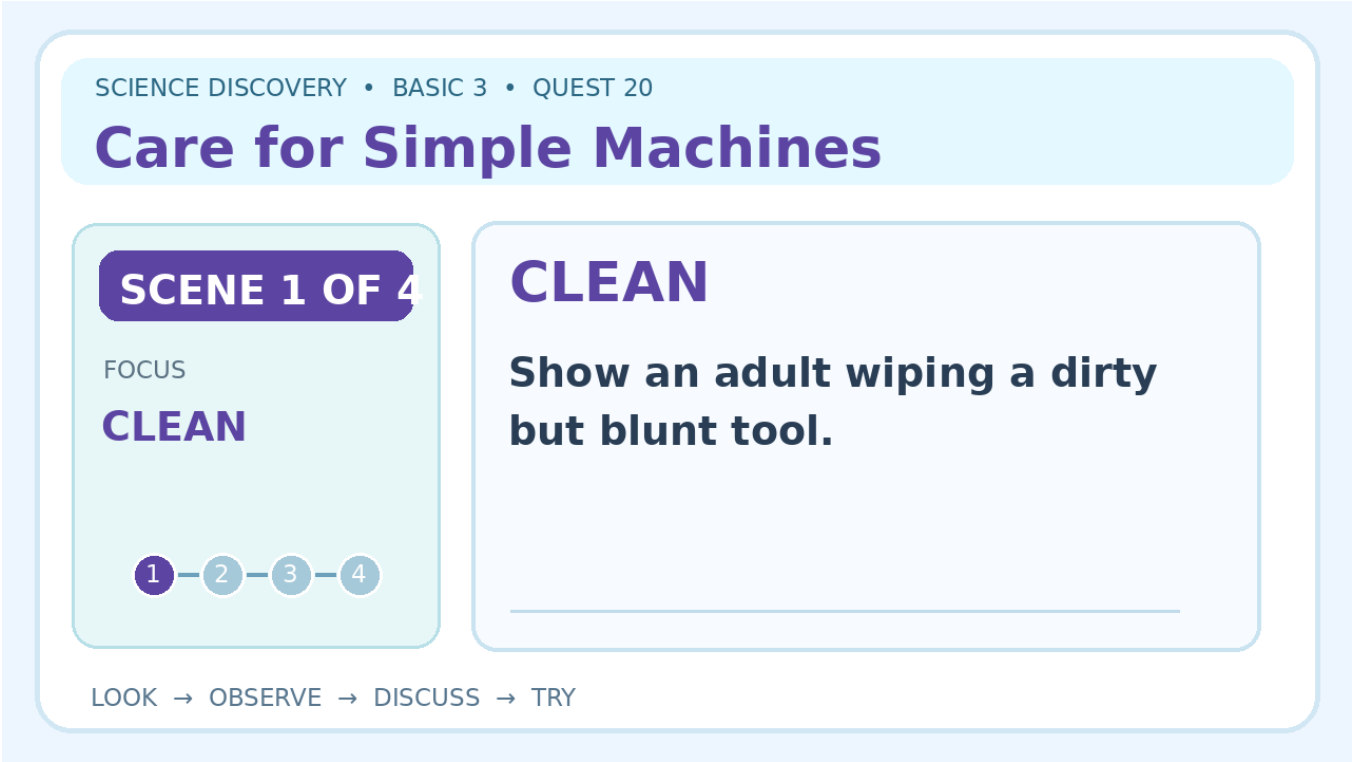
**Forces and Energy | Identify safe ways to clean, store and care for common simple machines and tools.**

## LET’S LEARN

Common tools such as scissors, bottle openers, hammers and ramps help people perform tasks. Proper care keeps tools useful and safer. Wipe dirt away, store tools in a dry suitable place and use them for their intended purpose. Adults may oil moving metal joints to help prevent rust or stiffness. Children should never handle sharp tools without supervision or try to repair dangerous machinery.

## Science words to remember

Cleaning • Storage • Lubrication • Safe use




## Try it for yourself

Look at pictures of a clean tool, a rusty tool and a tool in a dry cupboard. Sort safe and unsafe care actions. An adult can demonstrate wiping a blunt bottle opener.

Safety note: Only adults handle sharp blades, hammers, repairs or oiling demonstrations.

# QUEST 20: Show what you know

## Observe the four teaching steps

1. CLEAN: Show an adult wiping a dirty but blunt tool.

2. STORE: Show tools placed in a dry cabinet.

3. MAINTAIN: Show an adult putting a drop of oil on a metal hinge.

4. SAFE: Show protected storage of sharp tools out of child reach.

## Guided practice — answer in your exercise book

1. What helps keep a tool in good condition?
2. What should be removed after safe tool use?
3. Where should a metal tool be stored?
4. Which storage action helps reduce rust?
5. What may adults apply to a moving metal joint?
6. Why may a hinge need suitable oil?
7. How should scissors be used?
8. What is wise when a tool is damaged?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 21: Keep Our Environment Clean

**Humans and the Environment | Describe and practise ways to maintain a healthy school and home environment.**

## LET’S LEARN

A clean environment supports health and comfort. Put litter into bins, clean spills safely, separate waste when facilities allow and keep drains free from rubbish. Handwashing after cleaning is important. Sweeping may raise dust, so use appropriate protective measures and avoid breathing dust. Never pick up sharp, dirty or unknown objects. Group clean-up activities need adult guidance and appropriate materials.

## Science words to remember

Litter bins • Sweeping • Drainage • Handwashing




## Try it for yourself

Make a class clean-up poster with three safe steps. With adult supervision, put clean paper litter in a bin and wash hands afterwards.

Safety note: Never collect sharp waste or contaminated items. Protect against dust; adults organise clean-ups.

# QUEST 21: Show what you know

## Observe the four teaching steps

1. BIN: Show litter going into a covered or suitable bin.

2. SWEEP: Show a supervised careful dust-control clean-up.

3. DRAIN: Show a clear gutter compared with a blocked gutter.

4. WASH: Show soap-and-water handwashing after tidying.

## Guided practice — answer in your exercise book

1. Where should ordinary paper litter go?
2. What action helps keep a classroom tidy?
3. What may happen when sweeping a dusty room?
4. What can protect you from dust during clean-up?
5. Why should drains be kept clear of rubbish?
6. Which action helps prevent drain blockage?
7. What should you do after handling safe litter?
8. Why is handwashing useful after cleaning?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 22: Prevent Skin Problems

**Humans and the Environment | Describe hygienic habits that help reduce common skin infections and irritations.**

## LET’S LEARN

Skin helps protect our bodies. Washing regularly with clean water and mild soap can remove dirt and germs. Clean clothes, dry towels and not sharing personal towels can help lower the spread of some infections. Ringworm is a fungal skin infection; eczema is a skin condition that is not generally contagious. A new rash can have many causes. Tell a trusted adult and seek appropriate medical advice for persistent or concerning skin problems instead of trying unknown creams.

## Science words to remember

Washing • Clean clothing • Avoid sharing • Seek care




## Try it for yourself

Create a skin-care routine chart for washing, drying gently and using clean clothes. Discuss when a child should tell an adult about a rash.

Safety note: Do not apply unknown remedies or diagnose classmates by appearance.

# QUEST 22: Show what you know

## Observe the four teaching steps

1. BATHE: Show regular gentle washing with suitable soap.

2. CLOTHES: Show clean dry clothing and personal towels.

3. SHARE: Contrast personal items with shared classroom objects.

4. CARE: Show a child telling an adult about an unusual rash.

## Guided practice — answer in your exercise book

1. Which habit helps keep skin clean?
2. What should be used to clean dirty hands?
3. What clothing habit helps good skin hygiene?
4. What is sensible after bathing?
5. Which personal item is best not to share?
6. What can reduce some skin infection spread?
7. What should you do about a new painful skin rash?
8. Where can a caregiver get help for a persistent skin problem?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 23: Airborne Diseases

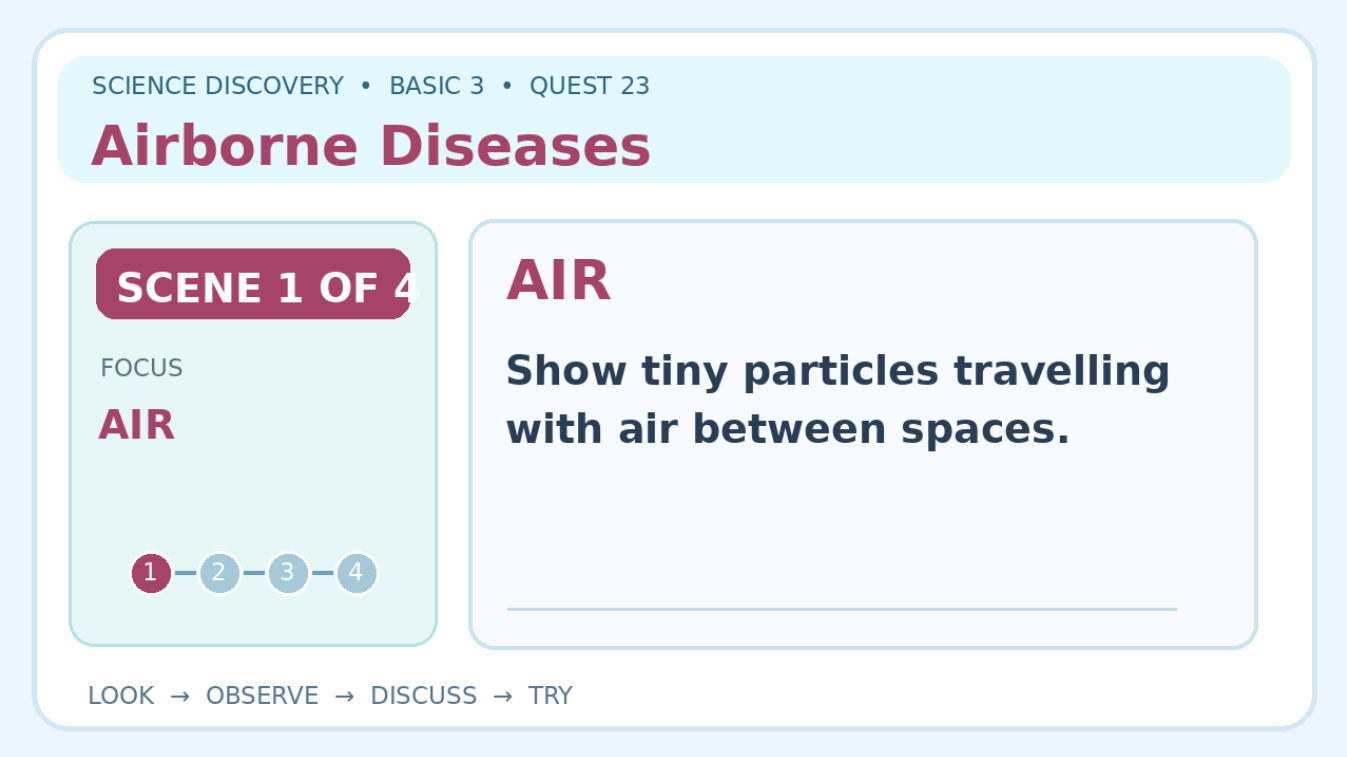
**Humans and the Environment | Explain that some illnesses spread through the air and identify protective habits.**

## LET’S LEARN

An airborne disease is an illness caused by germs that can spread through the air in small particles. Measles is one example. Some other respiratory infections spread through droplets and aerosols, including illnesses that cause coughs and colds. Coughing into an elbow or tissue, getting fresh airflow, handwashing and following recommended vaccinations can reduce infection risks. A cough is a symptom, not a disease by itself. Ask adults for help when someone is unwell.

## Science words to remember

Spread through air • Examples • Cough etiquette • Fresh air




## Try it for yourself

Practise covering a pretend cough with your elbow and make a classroom ventilation poster. Do not share personal medical stories about classmates.

Safety note: Do not deliberately expose anyone to coughs or sneezes; follow current medical guidance.

# QUEST 23: Show what you know

## Observe the four teaching steps

1. AIR: Show tiny particles travelling with air between spaces.

2. EXAMPLE: Show a label for measles as a disease, cough as a symptom.

3. COVER: Show a child coughing into the elbow.

4. VENTILATE: Show fresh air flowing through suitable classroom windows.

## Guided practice — answer in your exercise book

1. How can some infection-causing germs spread?
2. When might infectious respiratory particles enter air?
3. Which disease can spread through air?
4. Is a cough itself always a separate disease?
5. Where should you direct a cough if no tissue is available?
6. What should you do with a used tissue?
7. What can help improve fresh airflow in a room?
8. Which habit helps reduce the spread of respiratory infections?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 24: How Food Gets Spoiled

**Humans and the Environment | Identify observable signs and situations that can make foods unsafe to eat.**

## LET’S LEARN

Food can spoil when it is stored poorly, left too warm for too long, exposed to moisture or handled with dirty hands. Signs can include unusual smell, discolouration, visible mould or unexpected texture changes. Not all unsafe food looks or smells bad, so correct storage and use-by guidance matter. Keep covered food clean and store perishable foods correctly. Never taste suspected spoiled food to test it. This lesson focuses on observable spoilage, not naming microorganisms.

## Science words to remember

Appearance • Smell • Storage • Food safety




## Try it for yourself

Draw two food-storage scenes: covered food in a clean place and exposed food in a dirty hot place. Circle the safer storage choice.

Safety note: Never taste, sniff closely, touch or grow spoiled food in class; use pictures.

# QUEST 24: Show what you know

## Observe the four teaching steps

1. LOOK: Compare fresh bread and visibly mouldy bread (pictures only).

2. SMELL: Show a warning symbol for an unusual odour, no sniffing test.

3. STORE: Compare covered cool storage with exposed hot storage.

4. DECIDE: Mark suspected spoiled food as not safe to taste.

## Guided practice — answer in your exercise book

1. What visible change may show bread has spoiled?
2. What can be a warning sign on spoiled fruit?
3. What smell may warn that food has spoiled?
4. Should you taste food with a suspicious odour?
5. Which storage habit helps protect cooked food?
6. Where should food needing refrigeration be kept?
7. What should you do with suspected spoiled food?
8. Why are clean hands important when handling food?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# QUEST 25: Keep the Air Clean

**Humans and the Environment | Identify human actions that pollute the atmosphere and suggest safer alternatives.**

## LET’S LEARN

The atmosphere is the mixture of gases surrounding Earth. Some human activities make the air dirty, including burning rubbish, bush fires and smoke from vehicles or factories. Polluted air can bother breathing and harm the environment. Good choices include avoiding open burning, managing waste responsibly, maintaining vehicles and planting and protecting trees. This lesson is about recognising pollution and making practical improvements without frightening children.

## Science words to remember

Burning rubbish • Vehicle exhaust • Atmosphere • Prevention




## Try it for yourself

Draw two street scenes: one with rubbish burning near people and one with proper rubbish collection. Label which scene has cleaner air.

Safety note: Never burn waste as a class experiment or approach smoke. Use drawings only.

# QUEST 25: Show what you know

## Observe the four teaching steps

1. AIR: Show a clean-air neighbourhood with trees and homes.

2. POLLUTE: Show vehicle exhaust adding fumes to the air.

3. POLLUTE: Show unsafe burning of waste, with a stop symbol.

4. IMPROVE: Show waste bins and low-smoke, safer choices.

## Guided practice — answer in your exercise book

1. What can fill the air with harmful smoke near homes?
2. Which action is better than setting plastic waste on fire?
3. What may come from a poorly maintained vehicle exhaust?
4. Which action may help cut exhaust pollution?
5. What do we call the gases surrounding Earth?
6. Which statement describes the atmosphere?
7. What can help avoid smoke from waste?
8. Which school action supports cleaner air?

## Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

# Teacher and parent answer key

## Quest 1: Classifying Living Things

1. Plant • 2. A green plant • 3. Animals • 4. A dog • 5. It grows • 6. A lizard runs to shelter • 7. Animal group • 8. A plant

## Quest 2: Materials and Their Uses

1. It can be hard and withstand cooking heat • 2. A table • 3. It is flexible • 4. Plastic • 5. Clear glass • 6. A clear window • 7. Clothing • 8. Paper

## Quest 3: Objects Made from Several Materials

1. Wood and graphite • 2. Graphite core • 3. Glass • 4. Rubber • 5. Different parts need different properties • 6. It holds and protects the core • 7. Look without opening it • 8. Observe the outside and ask an adult

## Quest 4: Separate Solids from Liquids

1. A solid–liquid mixture • 2. The water • 3. Filtering through suitable filter material • 4. Undissolved sand • 5. Heavier particles may settle • 6. Near the bottom • 7. No • 8. Evaporation of water

## Quest 5: Changing States of Matter

1. It melts to liquid water • 2. Melting • 3. Freezing • 4. Removing enough heat • 5. Some water evaporates • 6. Water vapour • 7. Water vapour condenses • 8. Condensation

## Quest 6: Cycles in Daily Life

1. Day • 2. Day and night • 3. Wet and dry periods • 4. More frequent rain • 5. Ghana Independence Day • 6. A birthday • 7. 24 hours • 8. Once a year

## Quest 7: Why the Sun Matters

1. Natural light • 2. It helps us see • 3. Heat energy • 4. Seek shade and drink safe water • 5. To make food • 6. Green grass • 7. Wet clothes • 8. Drying harvested maize safely

## Quest 8: Rain, Snow, Hail and Sleet

1. Rain • 2. Rain drops • 3. Snow • 4. Snow • 5. Hail • 6. Hail • 7. Small ice pellets • 8. Sleet pellets

## Quest 9: What Makes Water Impure?

1. Litter thrown into it • 2. Dumping rubbish into it • 3. Pesticides • 4. Some mining waste • 5. Carry dirt and waste into it • 6. Waste into nearby water • 7. Keep rubbish and chemicals away • 8. Use safe treated water

## Quest 10: Air Is All Around Us

1. Air occupies space • 2. Air • 3. Yes • 4. It contains additional air with mass • 5. Wind • 6. A paper pinwheel • 7. Oxygen • 8. Open appropriate windows for ventilation

## Quest 11: Germinating Maize and Beans

1. On damp cotton wool • 2. Clean water • 3. They may not germinate • 4. To compare the effect of water • 5. A young root • 6. A young shoot • 7. Type of light and general temperature • 8. Record observations each day

## Quest 12: Body Parts Work Together

1. Eyes • 2. Eyes • 3. Ears • 4. With your ears • 5. Legs and feet • 6. Hands and arms • 7. Eyes and hands • 8. Ears, eyes and legs

## Quest 13: Sun, Earth and Moon

1. The Sun • 2. The Sun • 3. A planet • 4. The Sun • 5. The Moon • 6. Sunlight reflected by the Moon • 7. Earth rotating • 8. The Moon

## Quest 14: Homes for Living Things

1. Water in ponds or rivers • 2. Fins • 3. In a tree • 4. Wings • 5. Among grass and plants • 6. A grassy field • 7. They help it move through water • 8. It provides suitable resources and conditions

## Quest 15: Light as Energy

1. The Sun • 2. The Sun • 3. A torch • 4. An electric lamp • 5. Light from it reaches our eyes • 6. A suitable lamp • 7. A shadow • 8. On the side away from a light source

## Quest 16: What Makes Heat?

1. The Sun • 2. Energy from sunlight • 3. Friction • 4. Rubbing palms gently • 5. A stove • 6. An electric iron • 7. Ask an adult and keep away • 8. They can burn skin

## Quest 17: Everyday Uses of Heat

1. To prepare food • 2. Cooking rice • 3. Dry them when conditions are suitable • 4. A farmer • 5. An iron • 6. Tailoring or pressing clothes • 7. A blacksmith • 8. A baker

## Quest 18: Where Electricity Comes From

1. Dry-cell batteries • 2. A battery • 3. A solar panel • 4. Solar power • 5. Turn a turbine • 6. Moving water • 7. A closed battery torch • 8. Stay far away and tell an adult

## Quest 19: Forces Make Things Move

1. A push • 2. Pressing a door away from you • 3. Bringing a drawer toward you • 4. A pull • 5. A magnet • 6. Iron • 7. It stops moving • 8. Its speed or direction of movement

## Quest 20: Care for Simple Machines

1. Cleaning off dirt • 2. Dirt and moisture • 3. In a suitable dry place • 4. Keep tools dry • 5. Appropriate oil • 6. To reduce friction and stiffness • 7. For their intended cutting task with supervision • 8. Ask a responsible adult for help

## Quest 21: Keep Our Environment Clean

1. A suitable bin • 2. Putting rubbish in bins • 3. Dust can rise into the air • 4. Follow adult instructions and suitable dust protection • 5. Water can flow more safely • 6. Keep plastics out of drains • 7. Wash hands with soap and water • 8. It removes dirt and lowers infection risk

## Quest 22: Prevent Skin Problems

1. Bathing with clean water and suitable soap • 2. Soap and water • 3. Wear clean clothes regularly • 4. Dry skin gently with a clean towel • 5. A personal towel • 6. Using your own clean towel • 7. Tell a trusted adult • 8. From a qualified health professional

## Quest 23: Airborne Diseases

1. In particles through air • 2. When an infected person breathes or coughs • 3. Measles • 4. No, it is a symptom with different causes • 5. Into the bend of your elbow • 6. Dispose of it properly and clean hands • 7. Safe ventilation through suitable windows • 8. Follow health guidance and keep fresh air flowing

## Quest 24: How Food Gets Spoiled

1. Fuzzy mould patches • 2. Unexpected dark rotten patches • 3. An unusual sour or rotten odour • 4. No • 5. Keep it covered and stored safely • 6. In a suitable fridge • 7. Tell an adult and avoid eating it • 8. They lower contamination risk

## Quest 25: Keep the Air Clean

1. Burning rubbish • 2. Use proper waste collection • 3. Polluting fumes • 4. Proper vehicle maintenance • 5. The atmosphere • 6. Air around our planet • 7. Do not burn litter; manage it properly • 8. Keep rubbish out of fires and bins managed

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